

2021

HIGHER
SCHOOL
CERTIFICATE
TRIAL EXAMINATION

Mathematics Advanced

**General
Instructions**

- No reading time
- Working time – 2 hours
- Write using a black pen

**Total marks:
60**

Section I –10 marks

- Multiple choice questions
- Attempt Questions 1-10
- You have 20 minutes for this section
- You will not be able to move onto the next section until the time is up for this section

Section II –25 marks

- You have 50 minutes for this section
- You will not be able to move onto the next section until the time is up for this section

Section III –25 marks

- You have 50 minutes for this question

2021

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Advanced

Section I

10 marks

Attempt Questions 1-10

You have 20 minutes for this section

Instructions

- You will need to select your answers on the screen for this section.
 - Use the writing booklet for working only.
 - You will not be able to move onto the next section until the time is up for this section.
-

Section I

10 marks

Attempt questions 1-10

1 Given that $\log_a b = 3.75$ and $\log_a c = 0.25$, find $\log_a (bc)^2$.

- A. 0.879
- B. 0.9375
- C. 3.5
- D. 8

2 Which interval gives the range of the function $y = 6 \sin x - 10$?

- A. $[6, -10]$
- B. $[4, 16]$
- C. $[-16, -4]$
- D. $[-4, 16]$

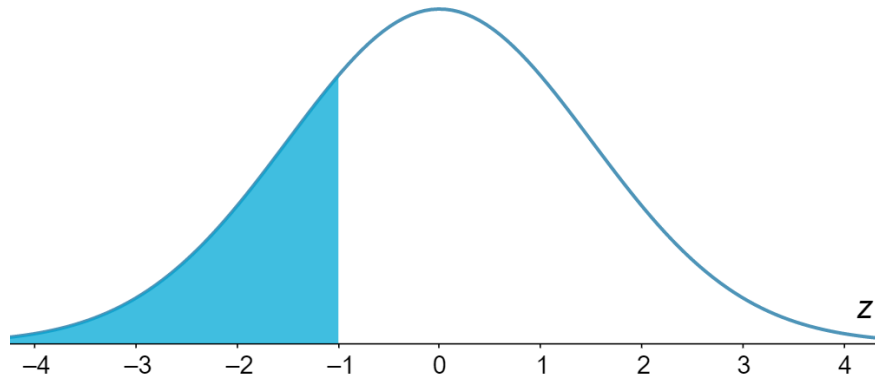
3 A random variable, T, has the following probability distribution:

t	$w-3$	$w-2$	$w-1$	w	$w+1$
$P(T=t)$	0.2	0.5	0.1	0.05	0.15

Given that $E(T)=8.45$, what is the value of w ?

- A. 2.69
- B. 3
- C. 6.9
- D. 10

- 4 The graph shows the area under a normal distribution curve.



Which of the following is equivalent to the shaded area?

- A. $P(z \geq 1)$
 - B. $P(z \geq -1)$
 - C. $1 - P(-1 \leq z \leq 1)$
 - D. $P(z \leq 1)$
- 5 If $f(x) = \sqrt{x}$ and $g(x) = 4 - x^2$, what is the domain of the function $f(g(x))$?

- A. $[-2, 2]$
- B. $(0, \infty)$
- C. $(-\infty, 2] \cup [2, \infty)$
- D. $(-\infty, \infty)$

- 6 Which expression is a term of the geometric series $2x - 6x^3 + 18x^5 - \dots$?

- A. $-1458x^{13}$
- B. $1458x^{13}$
- C. $-1458x^{15}$
- D. $1458x^{15}$

7 How many solutions does $2\sin^2 x - 1 = 0$ have for $-\frac{\pi}{2} \leq x \leq \pi$?

- A. 1
- B. 2
- C. 3
- D. 4

8 Which of the following is NOT a probability density function?

- A. $f(x) = \begin{cases} 3x^2 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{for all other values of } x \end{cases}$
- B. $f(x) = \begin{cases} 2x^4 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{for all other values of } x \end{cases}$
- C. $f(x) = \begin{cases} 4x^3 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{for all other values of } x \end{cases}$
- D. $f(x) = \begin{cases} 2x & \text{for } 0 \leq x \leq 1 \\ 0 & \text{for all other values of } x \end{cases}$

- 9 The equation of the tangent to the curve $y = f(x)$ at the point $(2, 7)$ is $y = 5x - 3$.

What is the equation of the tangent to the curve $f(x + 2) + 1$ at the point $(0, 8)$?

- A. $y = 3x - 4$
- B. $y = 5x - 13$
- C. $y = 5x - 12$
- D. $y = 5x + 8$

- 10 Tina either drinks tea or coffee in the morning. If she drinks tea one morning, the probability she drinks tea the next morning is 0.6. If she drinks coffee one morning, the probability she drinks coffee the next morning is 0.3.

Tina drank tea on a Monday morning.

What is the probability that she drinks coffee on the Wednesday morning?

- A. 0.12
- B. 0.24
- C. 0.36
- D. 0.40

End of Section I

2021

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Advanced

Section II

25 marks

You have 50 minutes for this section

Attempt Questions 11-18 (25 marks)

Instructions

- Answer the questions in the writing space provided
 - Write using black pen
 - You may answer the questions in any order. Just ensure your answer corresponds to the correct question number
 - You do not need to write the question
 - You will not be able to move onto the next section until the time is up for this section
-

Question 11 (2 marks)

A function $f(x)$ has the derivative $f'(x) = 4x^3 - 6x^2 + 5$ and $f(1) = 10$.

2

Find the function.

Question 12 (2 marks)

2

Find $\int \frac{x-3}{x^2-6x+5} dx$.

Question 13 (2 marks)

Explain whether or not the lowest score 3 is an outlier for the data set with 5-number summary.

2

Justify your answer with mathematical calculations.

L	Q_1	Q_2	Q_3	H
3	11	14	17	19

Question 14 (3 marks)

The graph of $y = \sqrt{x}$ undergoes the following sequence of transformations:

3

- Translation 3 units right
- Dilation horizontally by a factor of $\frac{1}{2}$
- Translation 1 unit down
- Dilation vertically by a factor of 4

Find the rule of the resulting graph.

Question 15 (4 marks)

(a) Show that $1 - \frac{4}{x-1} = \frac{x-5}{x-1}$.

1

(b) Hence, or otherwise, find $\int_2^e \frac{2x-10}{x-1} dx$, leave your answer in exact form.

3

Question 16 (4 marks)

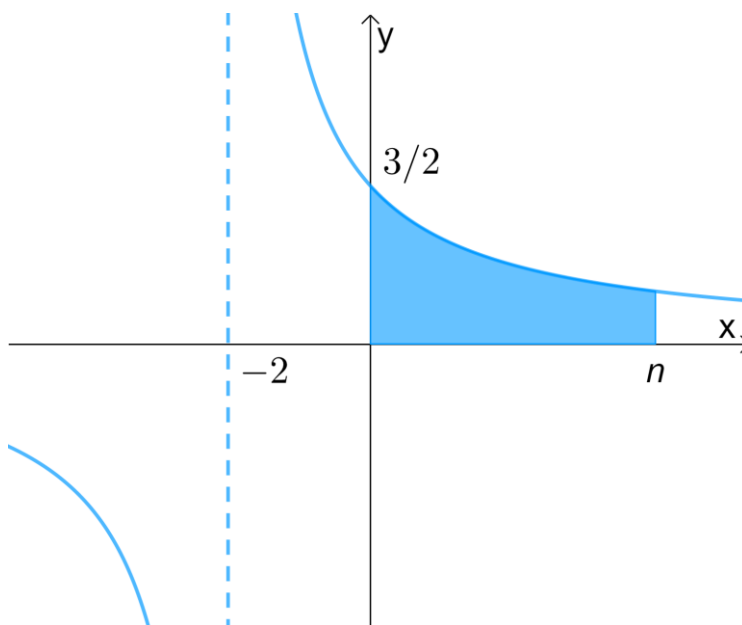
Sketch the graph of the curve $y = x(x^2 + 2x)^2$, labelling the stationary points and intercepts.

4

Question 17 (4 marks)

Part of the graph of the function $f(x) = \frac{a}{x+b}$ is given:

4



If the shaded area is $\ln 27$, find the values of a , b and n .

Question 18 (4 marks)

Boris makes yearly contributions of \$2000 into an investment fund. His investment earns 6% interest compounded annually.

4

After how many years will the total investment exceed \$45 000?

End of Section II

2021

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Advanced

Section III

25 marks

You have 50 minutes for this section

Attempt Questions 19-23 (25 marks)

Instructions

- Answer the questions in the writing space provided
 - Write using black pen
 - You may answer the questions in any order. Just ensure your answer corresponds to the correct question number
 - You do not need to write the question
-

Question 19 (2 marks)

Evaluate $\int_{-1}^4 (x - 3f(x)) dx$ given that $\int_{-1}^4 f(x) dx = 8$.

2

Question 20 (4 marks)

An object is moving in a straight line with acceleration function:

$$\ddot{x}(t) = -4te^{-t^2} \text{ m/s}^2$$

It is initially at the origin with a velocity of 1 m/s.

(a) Show that as t increases the object approaches a limiting velocity and state what the limiting velocity is.

2

(b) Determine the time when the particle is stationary.

2

Question 21 (8 marks)

A continuous random variable, X , has the following probability density function.

$$f(x) = \begin{cases} k \cos\left(\frac{\pi x}{2}\right) & \text{for } 0 \leq x \leq 1 \\ 0 & \text{for all other values of } x \end{cases}$$

(a) Show that $k = \frac{\pi}{2}$.

2

(b) Find $P\left(X \leq \frac{1}{3} \mid X \leq \frac{1}{2}\right)$.

3

(c) Find the median of X .

3

Question 22 (3 marks)

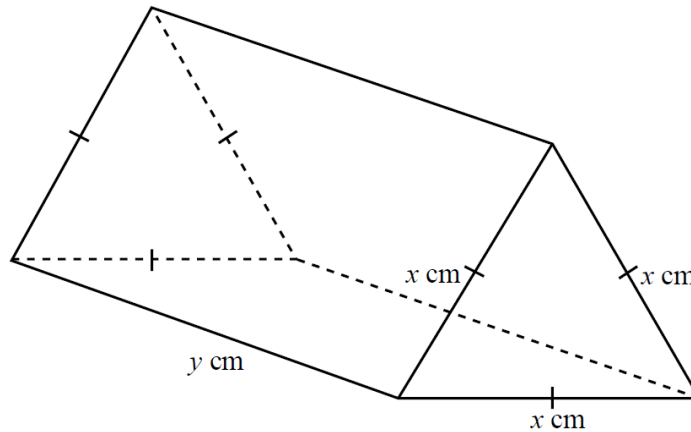
Solve $1 + \left(\frac{x+1}{x-1}\right) + \left(\frac{x+1}{x-1}\right)^2 + \left(\frac{x+1}{x-1}\right)^3 + \dots = \frac{x^2}{2}$. Leave your answer in exact form.

3

Question 23 (8 marks)

A gift box is made in the shape of a triangular prism.

The triangular end is an equilateral triangle with length x cm and the length of the box is y cm. The volume of the box is 300 cm^3 .



(a) Find an expression for y in terms of x .

2

(b) Show that the total surface area $A \text{ cm}^2$, of the box, is given by

2

$$A = \frac{1200\sqrt{3}}{x} + \frac{\sqrt{3}x^2}{2}.$$

(c) Find the value of x that gives the minimum total surface area. Justify your answer.

4

End of Paper

2021

HIGHER
SCHOOL
CERTIFICATE
TRIAL EXAMINATION

Mathematics Advanced

**General
Instructions**

- No reading time
- Working time – 2 hours
- Write using a black pen

**Total marks:
60**

Section I –10 marks

- Multiple choice questions
- Attempt Questions 1-10
- You have 20 minutes for this section
- You will not be able to move onto the next section until the time is up for this section

Section II –25 marks

- You have 50 minutes for this section
- You will not be able to move onto the next section until the time is up for this section

Section III –25 marks

- You have 50 minutes for this question

2021

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Advanced

Section I

10 marks

Attempt Questions 1-10

You have 20 minutes for this section

Instructions

- You will need to select your answers on the screen for this section.
 - Use the writing booklet for working only.
 - You will not be able to move onto the next section until the time is up for this section.
-

Section I

10 marks

Attempt questions 1-10

1 Given that $\log_a b = 3.75$ and $\log_a c = 0.25$, find $\log_a (bc)^2$.

A. 0.879

B. 0.9375

C. 3.5

☒ D. 8

2 Which interval gives the range of the function $y = 6 \sin x - 10$?

A. $[6, -10]$

B. $[4, 16]$

☒ C. $[-16, -4]$

D. $[-4, 16]$

3 A random variable, T , has the following probability distribution:

t	$w-3$	$w-2$	$w-1$	w	$w+1$
$P(T=t)$	0.2	0.5	0.1	0.05	0.15

Given that $E(T)=8.45$, what is the value of w ?

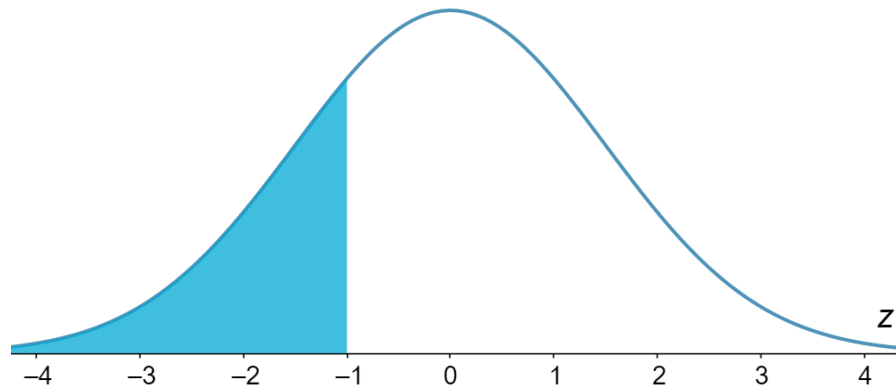
A. 2.69

B. 3

C. 6.9

☒ D. 10

- 4 The graph shows the area under a normal distribution curve.



Which of the following is equivalent to the shaded area?

- A. $P(z \geq 1)$
B. $P(z \geq -1)$
C. $1 - P(-1 \leq z \leq 1)$
D. $P(z \leq 1)$
- 5 If $f(x) = \sqrt{x}$ and $g(x) = 4 - x^2$, what is the domain of the function $f(g(x))$?

- A. $[-2, 2]$
B. $(0, \infty)$
C. $(-\infty, 2] \cup [2, \infty)$
D. $(-\infty, \infty)$

- 6 Which expression is a term of the geometric series $2x - 6x^3 + 18x^5 - \dots$?

- A. $-1458x^{13}$
B. $1458x^{13}$
C. $-1458x^{15}$
D. $1458x^{15}$

7 How many solutions does $2 \sin^2 x - 1 = 0$ have for $-\frac{\pi}{2} \leq x \leq \pi$?

A. 1

B. 2

☒ C. 3

D. 4

8 Which of the following is NOT a probability density function?

A. $f(x) = \begin{cases} 3x^2 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{for all other values of } x \end{cases}$

☒ B. $f(x) = \begin{cases} 2x^4 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{for all other values of } x \end{cases}$

C. $f(x) = \begin{cases} 4x^3 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{for all other values of } x \end{cases}$

D. $f(x) = \begin{cases} 2x & \text{for } 0 \leq x \leq 1 \\ 0 & \text{for all other values of } x \end{cases}$

- 9 The equation of the tangent to the curve $y = f(x)$ at the point $(2, 7)$ is $y = 5x - 3$.

What is the equation of the tangent to the curve $f(x + 2) + 1$ at the point $(0, 8)$?

- A. $y = 3x - 4$
- B. $y = 5x - 13$
- C. $y = 5x - 12$
- ☒ D. $y = 5x + 8$

- 10 Tina either drinks tea or coffee in the morning. If she drinks tea one morning, the probability she drinks tea the next morning is 0.6. If she drinks coffee one morning, the probability she drinks coffee the next morning is 0.3.

Tina drank tea on a Monday morning.

What is the probability that she drinks coffee on the Wednesday morning?

- A. 0.12
- B. 0.24
- ☒ C. 0.36
- D. 0.40

End of Section I

2021

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Advanced

Section II

25 marks

You have 50 minutes for this section

Attempt Questions 11-18 (25 marks)

Instructions

- Answer the questions in the writing space provided
 - Write using black pen
 - You may answer the questions in any order. Just ensure your answer corresponds to the correct question number
 - You do not need to write the question
 - You will not be able to move onto the next section until the time is up for this section
-

Question 11 (2 marks)

A function $f(x)$ has the derivative $f'(x) = 4x^3 - 6x^2 + 5$ and $f(1) = 10$.

2

Find the function.

$$\begin{aligned}
 f(x) &= \int 4x^3 - 6x^2 + 5 \, dx \\
 f(x) &= \frac{4x^4}{4} - \frac{6x^3}{3} + 5x + C \\
 f(x) &= x^4 - 2x^3 + 5x + C \quad (1) \\
 f(1) &= 10 \\
 10 &= (1)^4 - 2(1)^3 + 5(1) + C
 \end{aligned}$$

$C = 6$

$\therefore f(x) = x^4 - 2x^3 + 5x + 6 \quad (1)$

Question 12 (2 marks)

Find $\int \frac{x-3}{x^2-6x+5} \, dx$.

$$\begin{aligned}
 f(x) &= x^2 - 6x + 5 \\
 f'(x) &= 2x - 6 \\
 &= 2(x-3)
 \end{aligned}$$

2

$$\begin{aligned}
 &= \frac{1}{2} \int \frac{2(x-3)}{x^2-6x+5} \, dx \quad (1) \\
 &= \frac{1}{2} \ln(x^2-6x+5) + C \quad (1)
 \end{aligned}$$

Question 13 (2 marks)

Explain whether or not the lowest score 3 is an outlier for the data set with 5-number summary. Justify your answer with mathematical calculations.

2

L	Q_1	Q_2	Q_3	H
3	11	14	17	19

$$\begin{aligned}
 Q_1 - 1.5 \times IQR & \quad (1) \\
 &= 11 - 1.5 \times 6 \\
 &= 2 \\
 \therefore 2 &< 3 \quad \text{lowest score}
 \end{aligned}$$

$$\begin{aligned}
 IQR &= Q_3 - Q_1 \\
 &= 17 - 11 \\
 &= 6
 \end{aligned}$$

Hence 3 is not an outlier 1

Question 14 (3 marks)

The graph of $y = \sqrt{x}$ undergoes the following sequence of transformations:

3

- Translation 3 units right
- Dilation horizontally by a factor of $\frac{1}{2}$
- Translation 1 unit down
- Dilation vertically by a factor of 4

$$y = \sqrt{x-3}$$

$$y = \sqrt{2x-3} \quad (1)$$

$$y = \sqrt{2x-3} - 1 \quad (1)$$

$$y = 4\sqrt{2x-3} - 4 \quad (1)$$

OR

$$y = \sqrt{32x-48} - 4$$

Find the rule of the resulting graph.

Question 15 (4 marks)

(a) Show that $1 - \frac{4}{x-1} = \frac{x-5}{x-1}$.

1

(b) Hence, or otherwise, find $\int_2^e \frac{2x-10}{x-1} dx$, leave your answer in exact form.

3

$$\begin{aligned} (a) \quad \text{LHS} &= 1 - \frac{4}{x-1} \\ &= \frac{x-1}{x-1} - \frac{4}{x-1} \\ &= \frac{x-1-4}{x-1} \quad (1) \\ &= x-5 \\ &= \text{RHS} \end{aligned}$$

$$\begin{aligned} (b) \quad &\int_2^e \frac{2x-10}{x-1} dx \\ &= 2 \int_2^e \frac{x-5}{x-1} dx \\ &= 2 \int_2^e \left(1 - \frac{4}{x-1}\right) dx \\ &= 2 \int_2^e 1 dx - 8 \int_2^e \frac{1}{x-1} dx \quad (1) \\ &= 2 [x]_2^e - 8 [\ln(x-1)]_2^e \\ &= 2(e-2) - 8(\ln(e-1) - \ln(2-1)) \quad (1) \\ &= 2e-4 - 8\ln(e-1) \quad (1) \end{aligned}$$

Question 16 (4 marks)

Sketch the graph of the curve $y = x(x^2 + 2x)^2$, labelling the stationary points and intercepts.

4

$$u = x \quad v = (x^2 + 2x)^2$$

$$u' = 1 \quad v' = 2(x^2 + 2x) \cdot (2x + 2)$$

$$y' = 2x(2x+2)(x^2+2x) + (x^2+2x)^2$$

$$y' = (x^2+2x)(2x(2x+2) + (x^2+2x))$$

$$y' = x(x+2)(4x^2+4x+x^2+2x)$$

$$y' = x^2(x+2)(5x+6)$$

$$y' = 0 \rightarrow \text{Stat points}$$

$$0 = x^2(x+2)(5x+6)$$

$$\therefore \begin{pmatrix} x=0 \\ y=0 \end{pmatrix} \begin{pmatrix} x=-2 \\ y=0 \end{pmatrix} \begin{pmatrix} x=-\frac{6}{5} \\ y=-\frac{3456}{3125} \end{pmatrix}$$

hor. inf

①

max

①

min

①

$$\text{test } x = -2$$

$$x = -\frac{6}{5}$$

$$x = 0$$

x	-2.1	-2	-1.9
y'	+	0	-

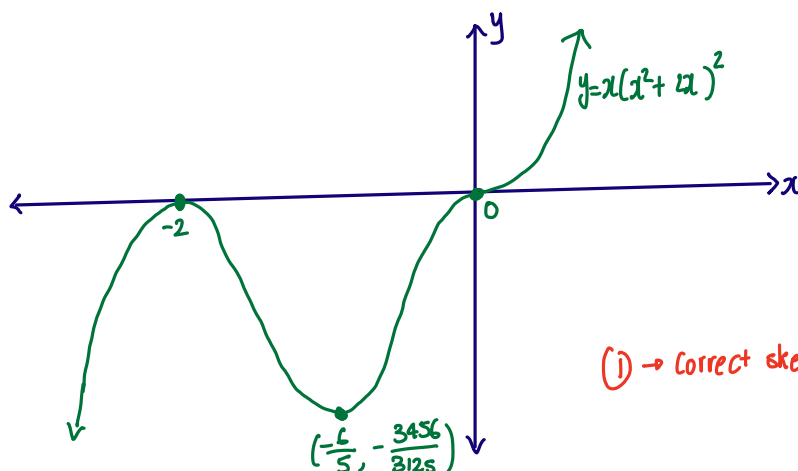
max

x	-1.3	$-\frac{6}{5}$	-1.1
y'	-	0	+

min

x	-0.1	0	0.1
y'	+	0	+

hor. pt. inf

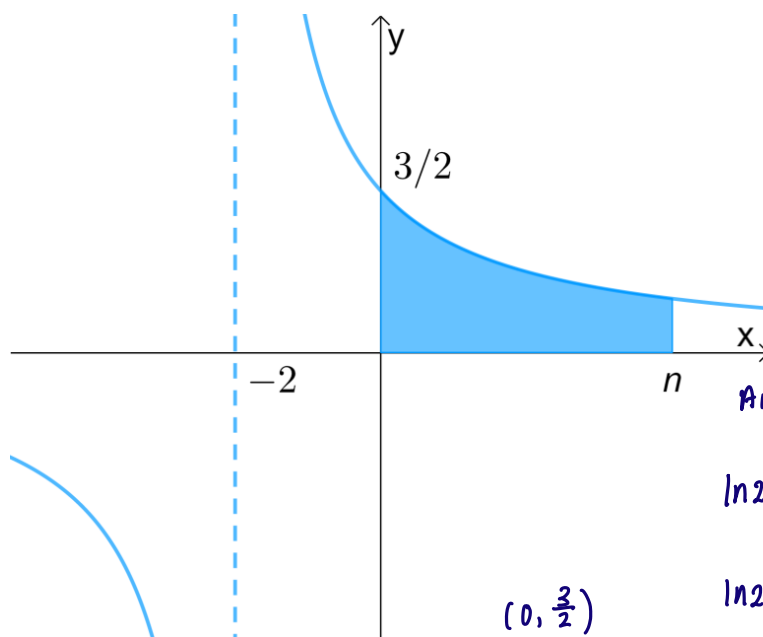


① → Correct sketch

Question 17 (4 marks)

Part of the graph of the function $f(x) = \frac{a}{x+b}$ is given:

4



$$\text{Area} = \ln 27$$

$$\ln 27 = \int_0^n \frac{3}{x+2} dx$$

$$\ln 27 = 3 \int_0^n \frac{1}{x+2} dx$$

$$\ln 27 = 3 [\ln(x+2)]_0^n \quad \text{①}$$

$$\ln 27 = 3 (\ln(n+2) - \ln(0+2))$$

$$\ln 27 = 3 \ln \frac{n+2}{2}$$

$$3 \ln 3 = 3 \ln \left(\frac{n+2}{2} \right)$$

$$\therefore 3 = \frac{n+2}{2}$$

$$6 = n+2$$

$$\therefore n = 4 \quad \text{①}$$

$$\left(0, \frac{3}{2}\right)$$

$$f(0) = \frac{3}{2}$$

$$\frac{3}{2} = \frac{a}{0+2}$$

$$\frac{3}{2} = \frac{a}{2}$$

$$\therefore a = 3 \quad \text{①}$$

If the shaded area is $\ln 27$, find the values of a , b and n .

$$\text{asym: } x+b=0$$

$$x = -2$$

$$b = 2 \quad \text{①}$$

Question 18 (4 marks)

Boris makes yearly contributions of \$2000 into an investment fund. His investment earns 6% interest compounded annually.

4

After how many years will the total investment exceed \$45 000?

$A_n = \text{value of the investment after } n \text{ years}$

$$A_1 = 2000 \times 1.06$$

$$A_2 = (A_1 + 2000) \cdot 1.06$$
$$= 2000 \cdot (1.06)^2 + 2000(1.06)$$

$$A_3 = (A_2 + 2000) \cdot 1.06$$

$$A_3 = 2000(1.06)^3 + 2000(1.06)^2 + 2000(1.06)$$

$\vdots$

$$A_n = 2000(1.06)^n + 2000(1.06)^{n-1} + \dots + 2000(1.06)$$

$$A_n = 2000(1.06^1 + \dots + 1.06^{n-1} + 1.06^n)$$

$$A_n = 2000 \cdot 1.06(1 + \dots + 1.06^{n-2} + 1.06^{n-1})$$

$$A_n = 2000 \times 1.06 \times \frac{1 \times (1.06^n - 1)}{1.06 - 1}$$

$$45000 = 2120 \times \frac{1.06^n - 1}{0.06} \quad (1)$$

$$1.273584906 = 1.06^n - 1$$

$$2.273584906 = 1.06^n$$

$$\frac{\ln 2.273584906}{\ln 1.06} = n$$

$$\therefore n = 14.09 \quad (1)$$

$$\therefore \text{It will take 15 years.} \quad (1)$$

End of Section II

2021

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Advanced

Section III

25 marks

You have 50 minutes for this section

Attempt Questions 19-23 (25 marks)

Instructions

- Answer the questions in the writing space provided
 - Write using black pen
 - You may answer the questions in any order. Just ensure your answer corresponds to the correct question number
 - You do not need to write the question
-

Question 19 (2 marks)

Evaluate $\int_{-1}^4 (x - 3f(x)) dx$ given that $\int_{-1}^4 f(x) dx = 8$.

2

$$= \int_{-1}^4 x dx - 3 \int_{-1}^4 f(x) dx$$

$$= \left[\frac{x^2}{2} \right]_{-1}^4 - 3 \times 8 \quad (1)$$

$$= \left(\frac{4^2}{2} - \frac{(-1)^2}{2} \right) - 24$$

$$= -\frac{33}{2} \quad (1)$$

Question 20 (4 marks)

An object is moving in a straight line with acceleration function:

$$\ddot{x}(t) = -4te^{-t^2} \text{ m/s}^2$$

It is initially at the origin with a velocity of 1 m/s.

(a) Show that as t increases the object approaches a limiting velocity and state what the limiting velocity is.

2

(b) Determine the time when the particle is stationary.

2

$$(a) \quad \dot{x} = \int -4te^{-t^2} dt$$

$$f(t) = -t^2 \quad f'(t) = -2t$$

$$\dot{x} = 2 \int (-2t \cdot e^{-t^2}) dt$$

$$\dot{x} = 2e^{-t^2} + C$$

$$\dot{x} = 1, t = 0$$

$$1 = 2e^{-0^2} + C$$

$$C = -1$$

$$\therefore \dot{x} = 2e^{-t^2} - 1$$

$$t \rightarrow \infty$$

$$\dot{x} = \frac{1}{2e^{t^2}} - 1$$

$$\dot{x} = \frac{1}{\infty} - 1$$

$$\dot{x} = -1$$

$$\therefore \text{as } t \rightarrow \infty$$

$$\dot{x} \rightarrow -1 \text{ m/s} \quad (1)$$

$$(b) \quad \text{particle stationary} \rightarrow \dot{x} = 0$$

$$t = ?$$

$$0 = 2e^{-t^2} - 1$$

$$1 = 2e^{-t^2}$$

$$1 = \frac{2}{e^{t^2}} \quad (1)$$

$$e^{t^2} = 2$$

$$t^2 = \ln 2$$

$$t = \pm \sqrt{\ln 2}$$

$$\therefore t = \sqrt{\ln 2} \quad t > 0 \quad (1)$$

Question 21 (8 marks)

A continuous random variable, X , has the following probability density function.

$$f(x) = \begin{cases} k \cos\left(\frac{\pi x}{2}\right) & \text{for } 0 \leq x \leq 1 \\ 0 & \text{for all other values of } x \end{cases}$$

- (a) Show that $k = \frac{\pi}{2}$. 2

$$1 = \int_0^1 k \cos\left(\frac{\pi x}{2}\right) dx$$

$$1 = \frac{2k}{\pi} \left[\sin\left(\frac{\pi x}{2}\right) \right]_0^1$$

$$1 = \frac{2k}{\pi} \left(\sin\frac{\pi}{2} - \sin 0 \right)$$

$$1 = \frac{2k}{\pi} (1 - 0)$$

$$\pi = 2k$$

$$\therefore k = \frac{\pi}{2} \quad (1)$$

- (b) Find $P\left(X \leq \frac{1}{3} \mid X \leq \frac{1}{2}\right)$. 3

$$= \frac{P\left(X \leq \frac{1}{3} \cap X \leq \frac{1}{2}\right)}{P\left(X \leq \frac{1}{2}\right)}$$

$$= \frac{\int_0^{\frac{1}{3}} \frac{\pi}{2} \cos\left(\frac{\pi x}{2}\right) dx}{\int_0^{\frac{1}{2}} \frac{\pi}{2} \cos\left(\frac{\pi x}{2}\right) dx} \quad (1)$$

$$= \frac{\left[\sin\left(\frac{\pi x}{2}\right) \right]_0^{\frac{1}{3}}}{\left[\sin\left(\frac{\pi x}{2}\right) \right]_0^{\frac{1}{2}}} \quad (1)$$

$$= \frac{\sin\frac{\pi}{6}}{\sin\frac{\pi}{4}}$$

$$= \frac{\frac{1}{2}}{\frac{\sqrt{2}}{2}}$$

$$= \frac{1}{\sqrt{2}} \quad (1)$$

- (c) Find the median of X . 3

median $\Rightarrow$

$$\frac{1}{2} = \int_0^m \frac{\pi}{2} \cos\left(\frac{\pi}{2} x\right) dx \quad (1)$$

$$\frac{1}{2} = \left[\sin\left(\frac{\pi}{2} \cdot x\right) \right]_0^m$$

$$\frac{1}{2} = \left[\sin\left(\frac{\pi}{2} \cdot m\right) - \sin\left(\frac{\pi}{2} \cdot 0\right) \right] \quad (1)$$

$$\frac{1}{2} = \sin \frac{\pi m}{2}$$

$$\therefore \frac{\pi m}{2} = \frac{\pi}{6}$$

$$\pi m = \frac{\pi}{3}$$

$$m = \frac{1}{3} \quad (1)$$

Question 22 (3 marks)

Solve $1 + \left(\frac{x+1}{x-1}\right) + \left(\frac{x+1}{x-1}\right)^2 + \left(\frac{x+1}{x-1}\right)^3 + \dots = \frac{x^2}{2}$. Leave your answer in exact form.

3

$$\begin{aligned}
 a &= 1 \\
 r &= \frac{x+1}{x-1} \\
 S_{\infty} &= \frac{a}{1-r} \\
 \frac{x^2}{2} &= \frac{1}{1 - \frac{x+1}{x-1}}
 \end{aligned}$$

$$\begin{aligned}
 \frac{x^2}{2} &= \frac{1}{\frac{x-1-x-1}{x-1}} \\
 \frac{x^2}{2} &= \frac{x-1}{-2} \\
 -2x^2 &= 2x-2 \\
 0 &= 2x^2+2x-2 \\
 0 &= x^2+x-1
 \end{aligned}$$

$$\begin{aligned}
 x &= \frac{-1 \pm \sqrt{1^2 - 4(1)(-1)}}{2} \\
 x &= \frac{-1 \pm \sqrt{5}}{2}
 \end{aligned}$$

Sum will only exist if $|r| < 1$

$$\left| \frac{x+1}{x-1} \right| < 1$$

when $x = \frac{-1+\sqrt{5}}{2}$ $\left| \frac{x+1}{x-1} \right| = 4.2... \neq 1$

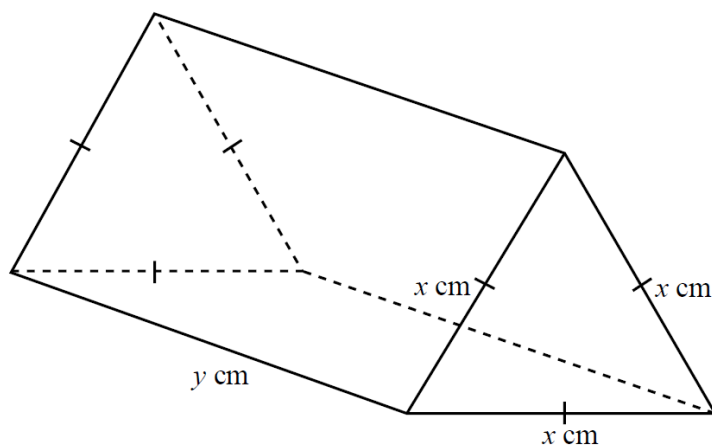
$x = \frac{-1-\sqrt{5}}{2}$ $\left| \frac{x+1}{x-1} \right| = 0.23... < 1$

Question 23 (8 marks)

A gift box is made in the shape of a triangular prism.

$$\therefore x = \frac{-1-\sqrt{5}}{2}$$

The triangular end is an equilateral triangle with length x cm and the length of the box is y cm. The volume of the box is 300 cm^3 .



(a) Find an expression for y in terms of x .

2

$$\begin{aligned}
 \text{Area of } \Delta &= \frac{1}{2}ab\sin C \\
 &= \frac{1}{2}x \cdot x \cdot \sin 60^\circ \\
 &= \frac{1}{2} \cdot x^2 \cdot \frac{\sqrt{3}}{2} \\
 &= \frac{\sqrt{3}x^2}{4}
 \end{aligned}$$

$$\begin{aligned}
 V &= A \times h \\
 300 &= \frac{\sqrt{3}}{4} x^2 \cdot y \\
 1200 &= \sqrt{3} x^2 \cdot y \\
 y &= \frac{1200}{\sqrt{3} x^2} \quad \text{or} \quad \frac{400\sqrt{3}}{x^2}
 \end{aligned}$$

- (b) Show that the total surface area $A \text{ cm}^2$, of the box, is given by

2

$$A = \frac{1200\sqrt{3}}{x} + \frac{\sqrt{3}x^2}{2}.$$

$$SA = 3 \times \square + 2 \times \triangle$$

$$= 3xy + 2 \times \frac{\sqrt{3}}{4} x^2 \quad (1)$$

$$= 3x \cdot \frac{400\sqrt{3}}{x^2} + \frac{2\sqrt{3}x^2}{4}$$

$$= \frac{1200\sqrt{3}}{x} + \frac{\sqrt{3}x^2}{2} \quad (1)$$

- (c) Find the value of x that gives the minimum total surface area. Justify your answer.

4

$$S = 1200\sqrt{3}x^{-1} + \frac{\sqrt{3}}{2}x^2 \quad \frac{dS}{dx} = -1200\sqrt{3}x^{-2} + \sqrt{3}x$$

$$\frac{dS}{dx} = \frac{-1200\sqrt{3}}{x^2} + \sqrt{3}x \quad (1)$$

$$\frac{dS}{dx} = 0 \rightarrow \text{stat points}$$

$$0 = \frac{-1200\sqrt{3}}{x^2} + \frac{\sqrt{3}x^3}{x^2}$$

$$0 = -1200\sqrt{3} + \sqrt{3}x^3$$

$$1200\sqrt{3} = \sqrt{3}x^3 \quad (1)$$

$$x = \sqrt[3]{1200}$$

$$\frac{d^2S}{dx^2} = \frac{2400\sqrt{3}}{x^3} + \sqrt{3} \quad (1)$$

$$\frac{d^2S}{dx^2} (\sqrt[3]{1200}) = \frac{2400\sqrt{3}}{1200} + \sqrt{3}$$

$$= 3\sqrt{3} > 0 \quad \checkmark \text{ min}$$

$$\therefore x = \sqrt[3]{1200} \text{ gives minimum SA.} \quad (1)$$

End of Paper